

## Appendix A: BSA-D2 Hardware and Actuator Settings

| Parameter                      | Value        | Unit                         |
|--------------------------------|--------------|------------------------------|
| BSA-D2 mass                    | 19.450606146 | kg                           |
| COM $x$                        | 0.048901198  | m                            |
| COM $y$                        | 0.020260562  | m                            |
| COM $z$                        | 0.048221021  | m                            |
| Inertia $J_{xx}$               | 0.193749459  | $\text{kg} \cdot \text{m}^2$ |
| Inertia $J_{xy}$               | 0.017485528  | $\text{kg} \cdot \text{m}^2$ |
| Inertia $J_{xz}$               | 0.033545050  | $\text{kg} \cdot \text{m}^2$ |
| Inertia $J_{yy}$               | 1.847824640  | $\text{kg} \cdot \text{m}^2$ |
| Inertia $J_{yz}$               | 0.006924042  | $\text{kg} \cdot \text{m}^2$ |
| Inertia $J_{zz}$               | 1.895445823  | $\text{kg} \cdot \text{m}^2$ |
| Action dimension               | 5            | –                            |
| Action lower bound             | –1           | –                            |
| Action upper bound             | 1            | –                            |
| Forward-force scale            | –90          | N                            |
| Lateral-force scale            | –60          | N                            |
| Vertical-force scale           | 45           | N                            |
| Pitch-torque scale             | 18           | $\text{N} \cdot \text{m}$    |
| Yaw-torque scale               | 18           | $\text{N} \cdot \text{m}$    |
| Roll PD $k_p$                  | 6.0          | –                            |
| Roll PD $k_d$                  | 3.0          | –                            |
| Roll torque limit              | 8.0          | $\text{N} \cdot \text{m}$    |
| Allocation force fraction      | 0.50         | –                            |
| Allocation warning threshold   | $10^{-3}$    | –                            |
| Vehicle thruster reverse limit | –46          | N                            |
| Vehicle thruster forward limit | 57           | N                            |
| PWM neutral value              | 1500         | $\mu\text{s}$                |
| Flow-bank thruster count       | 12           | thrusters                    |
| Flow-bank per-thruster bound   | 80           | N                            |
| Flow-bank aggregate ceiling    | 960          | N                            |

| Allocation matrix | $T_1$   | $T_2$  | $T_3$   | $T_4$   | $T_5$   | $T_6$   |
|-------------------|---------|--------|---------|---------|---------|---------|
| $F_x$             | 0.000   | 1.000  | –1.000  | 0.000   | 0.000   | 0.000   |
| $F_y$             | –0.866  | 0.000  | 0.000   | 0.500   | –0.866  | 0.500   |
| $F_z$             | –0.500  | 0.000  | 0.000   | –0.866  | –0.500  | –0.866  |
| $M_x$             | 0.1497  | 0.0000 | 0.0000  | 0.0570  | –0.0761 | –0.1688 |
| $M_y$             | 0.1828  | 0.1011 | –0.0184 | –0.2584 | –0.1492 | 0.3167  |
| $M_z$             | –0.3167 | 0.1242 | 0.1843  | –0.1492 | 0.2584  | 0.1828  |

## Appendix B: Isaac Sim and OceanSim Simulation Parameters

| Parameter                         | Value                | Unit   | Source/Use |
|-----------------------------------|----------------------|--------|------------|
| Physics rate                      | 50                   | Hz     | PhysX      |
| Policy rate                       | 10                   | Hz     | SAC        |
| Decimation                        | 5                    | –      | PhysX/SAC  |
| Mass randomization range          | [16, 22]             | kg     | random.    |
| Inertia-scale randomization range | [0.60, 1.60]         | –      | random.    |
| Linear damping $D_1$              | diag(8.0, 8.0, 6.0)  | –      | drag       |
| Quadratic damping $D_2$           | diag(4.0, 4.0, 3.0)  | –      | drag       |
| Training current speeds           | {0.1, 0.3, 0.5}      | m/s    | train      |
| Evaluation current speeds         | {0.0, 0.1, 0.3, 0.5} | m/s    | eval.      |
| Training current directions       | {0°, ±90°}           | –      | train      |
| Evaluation current directions     | {0°, ±90°}           | –      | eval.      |
| $B_{\infty,R}$                    | 0.02                 | –      | OceanSim   |
| $B_{\infty,G}$                    | 0.36                 | –      | OceanSim   |
| $B_{\infty,B}$                    | 0.32                 | –      | OceanSim   |
| $\beta_{bs,R}$                    | 0.18                 | –      | OceanSim   |
| $\beta_{bs,G}$                    | 0.10                 | –      | OceanSim   |
| $\beta_{bs,B}$                    | 0.06                 | –      | OceanSim   |
| $\beta_{attn,R}$                  | 0.35                 | –      | OceanSim   |
| $\beta_{attn,G}$                  | 0.20                 | –      | OceanSim   |
| $\beta_{attn,B}$                  | 0.12                 | –      | OceanSim   |
| $s_{attn}$                        | 0.20                 | –      | OceanSim   |
| $s_{bs}$                          | 0.35                 | –      | OceanSim   |
| $k_I$                             | 1.80                 | –      | OceanSim   |
| $\gamma$                          | 0.75                 | –      | OceanSim   |
| Native camera width               | 640                  | pixels | camera     |
| Native camera height              | 360                  | pixels | camera     |
| Near clipping distance            | 0.10                 | m      | camera     |
| Far clipping distance             | 180.0                | m      | camera     |
| Camera translation                | (0.80, 0.00, 0.05)   | m      | camera     |
| Camera quaternion                 | (1, 0, 0, 0)         | –      | camera     |
| Policy input width                | 256                  | pixels | RGB-D      |
| Policy input height               | 144                  | pixels | RGB-D      |
| Minimum valid depth               | 0.25                 | m      | depth      |
| Sector depth quantile             | 0.10                 | –      | clearance  |
| Minimum valid-pixel ratio         | 0.30                 | –      | depth      |
| Visual update interval            | 2                    | steps  | visual     |
| Visual refresh rate               | 5                    | Hz     | visual     |
| Visual observation dimension      | 31                   | –      | obs.       |

## Appendix B (continued): Underwater Image Formation

Under the adopted OceanSim underwater image-formation model and parameterization, the degraded RGB channel  $c \in \{R, G, B\}$  is

$$\begin{aligned}\tilde{I}_{t,c} = & I_{0,t,c} \exp(-d_t \beta_{\text{attn},c} s_{\text{attn}}) \\ & + B_{\infty,c} s_{\text{bs}} [1 - \exp(-d_t \beta_{\text{bs},c} s_{\text{attn}})],\end{aligned}\tag{B.1}$$

followed by the intensity and gamma corrections

$$I_{t,c} = \text{clip}\left(k_I \tilde{I}_{t,c}, 0, 1\right)^\gamma.\tag{B.2}$$

Here,  $I_{0,t,c}$  and  $d_t$  denote the normalized RGB and depth images, respectively;  $\beta_{\text{attn},c}$ ,  $\beta_{\text{bs},c}$ , and  $B_{\infty,c}$  are the channel-dependent attenuation, backscattering, and veiling-light parameters; and  $s_{\text{attn}}$ ,  $s_{\text{bs}}$ ,  $k_I$ , and  $\gamma$  scale the exponential terms, veiling-light amplitude, intensity, and gamma correction, respectively. All operations are pixelwise. The corresponding parameters are listed in the preceding table.

## Appendix C: CSPI-LLM Component and Network Architecture

| Item                           | Value                                                                                                                                           | Use                             |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| Teacher window                 | $(t - 1.5, t - 1.0, t - 0.5, t)$                                                                                                                | 4-frame window                  |
| Window rate                    | 2 Hz                                                                                                                                            | from 10 Hz trajectory, stride 5 |
| Allowed inputs                 | body/DVL velocity, IMU angular rate, body-frame goal direction, goal-distance trend, directional clearances, previous action, Fossen-lite proxy | deployable observation          |
| Excluded inputs                | route ID, true current, relative-water truth, waypoint/cross-track oracle, oracle progress, reward terms                                        | no privileged input             |
| Lateral drift proxy            | $d_{y,t} = \hat{v}_t - 0.25a_{F_y,t-1}$                                                                                                         | Fossen-lite                     |
| Vertical drift proxy           | $d_{z,t} = \hat{w}_t - 0.25a_{F_z,t-1}$                                                                                                         | Fossen-lite                     |
| Drag proxy                     | $d_{D,t} = \ \hat{b}\hat{\mathbf{v}}_t \odot \ \hat{b}\hat{\mathbf{v}}_t\ _2\ _2$                                                               | Fossen-lite                     |
| Progress-loss proxy            | $d_{\text{loss},t} = \max(0, p_{\text{expected},t} - p_{\text{observed},t})$                                                                    | Fossen-lite                     |
| Clearance margin               | $d_i - d_{\text{safe},i}$                                                                                                                       | safety evidence                 |
| $a_{\text{dec}}^{\text{cons}}$ | 0.50 m/s <sup>2</sup>                                                                                                                           | safety-distance/barrier         |
| Safety thresholds              | front 12.0 m, side 8.0 m, vertical 8.0 m                                                                                                        | Fossen-lite                     |
| Bin thresholds                 | lateral 1.50 m/s, vertical 1.30 m/s, progress-loss 0.30 m/s, drag 2.25 m <sup>2</sup> /s <sup>2</sup>                                           | qualitative evidence            |
| Output schema                  | 8 yes/no keys                                                                                                                                   | teacher target                  |
| Reserved ninth field           | STOP_HOLD = 0                                                                                                                                   | semantic vector                 |
| Final filtering                | 48,599 $\rightarrow$ 48,587                                                                                                                     | SFT dataset                     |
| Removed samples                | empty 4, left-right conflict 4, up-down conflict 4                                                                                              | filtering                       |
| Semantic executability         | left/right and up/down are mutually exclusive                                                                                                   | not formal reachability         |
| Online safety fusion           | front_critical 0.08, front_low 0.16, side_critical 0.08, side_low 0.14, reverse_critical 0.04, direction_margin 0.08                            | semantic label correction       |
| Safety-fusion scope            | modifies semantic labels only; does not overwrite SAC action                                                                                    | online fallback                 |



| Component                     | Parameter                          | Value                |
|-------------------------------|------------------------------------|----------------------|
| LLM checkpoint                | Local recovered backbone           | Qwen3-4B-Instruction |
| LLM text stack                | Hidden layers                      | 36                   |
| LLM text stack                | Hidden size                        | 2560                 |
| LLM text stack                | Attention heads                    | 32                   |
| LLM text stack                | Key-value heads                    | 8                    |
| LLM text stack                | Intermediate size                  | 9728                 |
| LLM text stack                | Maximum positions                  | 40960                |
| LLM text stack                | Vocabulary size                    | 151936               |
| Vision transformer block      | Hidden size                        | 768                  |
| Vision transformer block      | Intermediate size                  | 3072                 |
| Vision transformer block      | Output hidden size                 | 1024                 |
| Semantic output head          | Multilabel semantic vector         | 9 labels             |
| Semantic gate input           | Semantic vector plus hold age      | 10 values            |
| Visual branch                 | Four-frame RGB-D history embedding | $d_v = 128$          |
| Visual stream Transformer     | Layers                             | 2                    |
| Visual stream Transformer     | Attention heads                    | 4                    |
| Visual stream Transformer     | $d_{\text{model}}$                 | 128                  |
| Navigation branch             | Deployable navigation embedding    | $d_n = 64$           |
| Navigation stream Transformer | Layers                             | 2                    |
| Navigation stream Transformer | Attention heads                    | 4                    |
| Navigation stream Transformer | $d_{\text{model}}$                 | 64                   |
| Feature fusion                | Fused feature dimension            | $d_f = 256$          |
| Fusion encoder                | Layers                             | 2                    |
| Fusion encoder                | Attention heads                    | 4                    |
| Fusion encoder                | $d_{\text{model}}$                 | 256                  |
| Policy input                  | State dimension                    | 47 values            |
| Actor head                    | Action dimension                   | 5                    |
| Actor distribution            | Squashed Gaussian policy           | enabled              |
| Actor log-std range           | Lower / upper bounds               | $[-5, 1]$            |
| Critic heads                  | Twin Q critics                     | 2 critics            |

## Appendix D: LLM and DRL Training Hyperparameters

| Parameter                 | Value                | Unit        | Source/Use |
|---------------------------|----------------------|-------------|------------|
| Filtered semantic samples | 48587                | samples     | LLM train  |
| LoRA rank $r_L$           | 8                    | –           | LLM train  |
| LoRA scale $\alpha_L$     | 16                   | –           | LLM train  |
| LoRA dropout              | 0.05                 | –           | LLM train  |
| LLM learning rate         | $1.5 \times 10^{-5}$ | –           | LLM train  |
| Warmup ratio              | 0.05                 | –           | LLM train  |
| Per-device batch          | 8                    | samples     | LLM train  |
| Gradient accumulation     | 2                    | steps       | LLM train  |
| Effective batch           | 16                   | samples     | LLM train  |
| LLM epochs                | 2.0                  | epochs      | LLM train  |
| Sequence length           | 1024                 | tokens      | LLM train  |
| Validation split          | 0.05                 | –           | LLM train  |
| Evaluation batch          | 8                    | samples     | LLM train  |
| $\gamma_{\text{SAC}}$     | 0.995                | –           | SAC train  |
| SAC learning rate         | $3.0 \times 10^{-4}$ | –           | SAC train  |
| Target smoothing $\tau$   | 0.005                | –           | SAC train  |
| Replay buffer size        | 300000               | transitions | SAC train  |
| Learning start            | 50000                | transitions | SAC train  |
| SAC batch size            | 512                  | samples     | SAC train  |
| Train frequency           | 1                    | step        | SAC train  |
| Gradient steps            | 1                    | step        | SAC train  |
| Entropy coefficient       | auto_0.05            | –           | SAC train  |
| Target entropy            | −5.0                 | –           | SAC train  |
| Target update interval    | 1                    | step        | SAC train  |
| Optimizer epsilon         | $10^{-5}$            | –           | SAC train  |

## Appendix D (continued): Reward and Training Objectives

The terminal reward replaces the nonterminal progress, time, action-magnitude, and action-variation reward on terminal transitions:

$$r_t^{\text{term}} = \begin{cases} +300, & \text{success,} \\ -300, & \text{contact,} \\ -300, & \text{out of bounds,} \\ -100, & \text{timeout,} \\ 0, & \text{otherwise.} \end{cases} \quad (\text{D.1})$$

The nonterminal weights are  $w_d = 8.0$ ,  $w_t = -0.05$ ,  $w_a = 0.01$ , and  $w_{\Delta a} = 0.005$ , with initial previous action  $\mathbf{a}_{-1} = \mathbf{0}$ . The safe-distance progress field is unavailable to the actor and both language models. For a frozen weight matrix  $\mathbf{W}_0$ , an adapted layer is

$$\mathbf{h} = \mathbf{W}_0 \mathbf{x} + \frac{\alpha_L}{r_L} \mathbf{B} \mathbf{A} \mathbf{x}, \quad (\text{D.2})$$

where  $\mathbf{x}$  and  $\mathbf{h}$  denote the layer input and output,  $\mathbf{A}$  and  $\mathbf{B}$  are the trainable low-rank factors, and  $r_L$  and  $\alpha_L$  are the LoRA rank and scale. Rank, scale, and dropout are listed in the preceding hyperparameter table.

The actor and critic interfaces are defined in Section III-D of the main text. Here,  $\tilde{\mathbf{z}}_t^\pi$  and  $\tilde{\mathbf{z}}_t^{Q_i}$  denote their independently gated features, while  $\theta$  and  $\phi_i$  parameterize the actor and the  $i$ th critic.

The SAC target is

$$y_t = r_t + \gamma_{\text{RL}}(1 - d_t) \left[ \min_i Q_{\bar{\phi}_i}(\tilde{\mathbf{z}}_{t+1}^{Q_i}, \mathbf{a}_{t+1}) - \alpha \log \pi_\theta(\mathbf{a}_{t+1} \mid \tilde{\mathbf{z}}_{t+1}^\pi) \right], \quad (\text{D.3})$$

where  $d_t$  is the terminal indicator,  $\alpha$  is the entropy coefficient, and  $\bar{\phi}_i$  denotes the target-critic parameters. The critic and actor objectives are

$$\begin{aligned} J_Q(\phi_i) &= \mathbb{E} \left[ \left( Q_{\phi_i}(\tilde{\mathbf{z}}_t^{Q_i}, \mathbf{a}_t) - y_t \right)^2 \right], \\ J_\pi(\theta) &= \mathbb{E} \left[ \alpha \log \pi_\theta(\mathbf{a}_t \mid \tilde{\mathbf{z}}_t^\pi) - \min_i Q_{\phi_i}(\tilde{\mathbf{z}}_t^{Q_i}, \mathbf{a}_t) \right]. \end{aligned} \quad (\text{D.4})$$

The target and objectives retain the standard SAC formulation. Semantic conditioning is supplied through the independently gated actor and critic features defined in the main text.

## Appendix E: LLM Internal Component Comparison

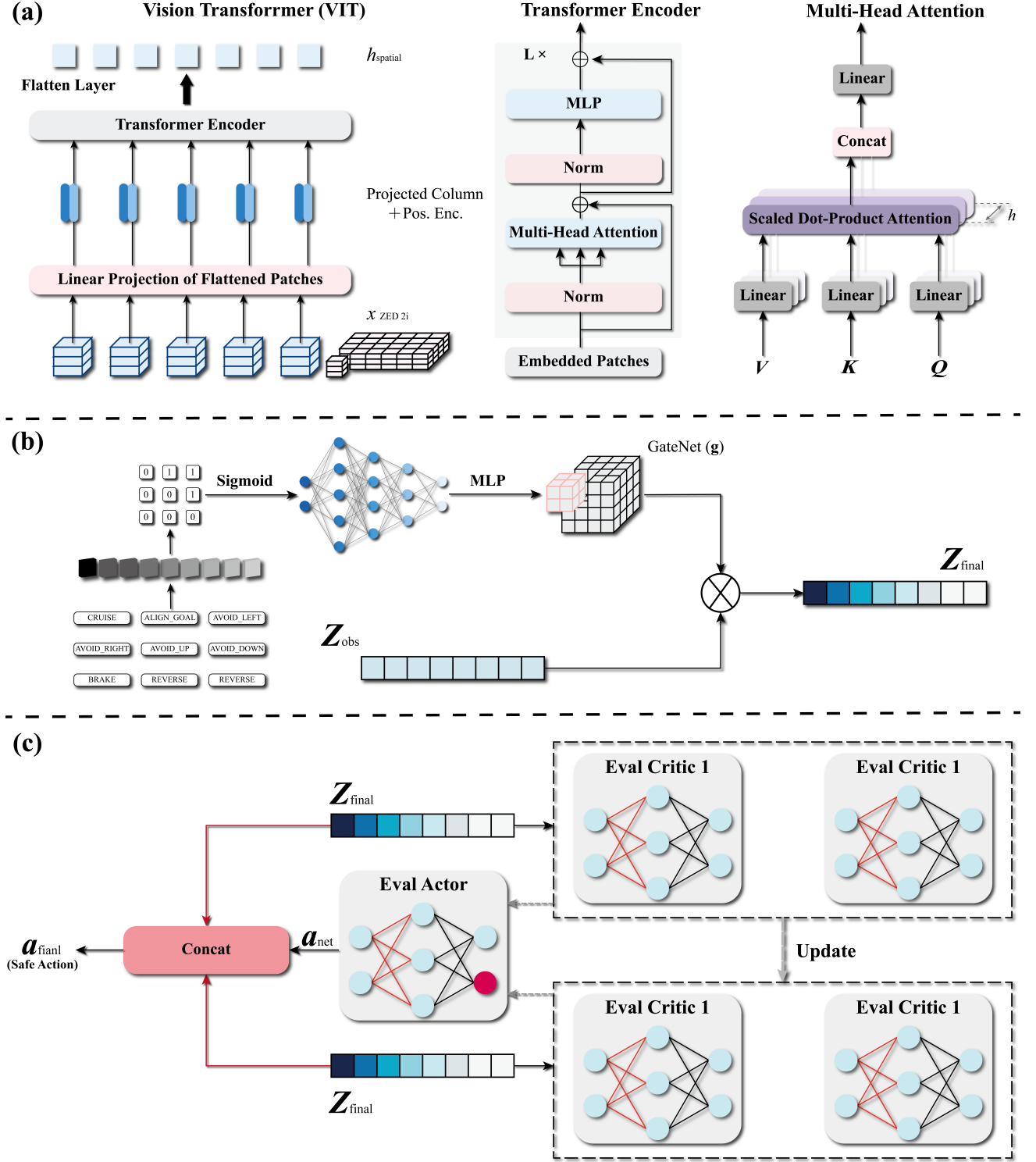
| Component / field          | Qwen3-0.6B              | Qwen3-4B-Instruction    | Qwen3-8B                |
|----------------------------|-------------------------|-------------------------|-------------------------|
| Family                     | Qwen3                   | Qwen3                   | Qwen3                   |
| Model class                | Qwen3ForCausalLM        | Qwen3ForCausalLM        | Qwen3ForCausalLM        |
| Model type                 | qwen3                   | qwen3                   | qwen3                   |
| Parameter scale            | 0.6B                    | 4B                      | 8B                      |
| Architecture type          | decoder-only causal LM  | decoder-only causal LM  | decoder-only causal LM  |
| Hidden size                | 1024                    | 2560                    | 4096                    |
| Decoder layers             | 28                      | 36                      | 36                      |
| Attention heads            | 16                      | 32                      | 32                      |
| Key-value heads            | 8                       | 8                       | 8                       |
| Head dimension             | 128                     | 128                     | 128                     |
| Attention projection width | 2048                    | 4096                    | 4096                    |
| Query heads per KV head    | 2                       | 4                       | 4                       |
| GQA setting                | grouped-query attention | grouped-query attention | grouped-query attention |
| MLP intermediate size      | 3072                    | 9728                    | 12288                   |
| FFN expansion ratio        | 3.00                    | 3.80                    | 3.00                    |
| Activation                 | SiLU                    | SiLU                    | SiLU                    |
| Vocabulary size            | 151936                  | 151936                  | 151936                  |
| Token embedding shape      | $151936 \times 1024$    | $151936 \times 2560$    | $151936 \times 4096$    |
| Embedding parameters       | 155.6M                  | 389.0M                  | 622.3M                  |
| Output embedding tying     | tied                    | tied                    | untied                  |
| Maximum positions          | 40960                   | 40960                   | 40960                   |
| Maximum window layers      | 28                      | 36                      | 36                      |
| Sliding-window size        | null                    | null                    | null                    |
| Use sliding window         | false                   | false                   | false                   |
| RMSNorm epsilon            | $10^{-6}$               | $10^{-6}$               | $10^{-6}$               |
| RoPE theta                 | 1000000                 | 1000000                 | 1000000                 |
| RoPE scaling               | null                    | null                    | null                    |
| Initializer range          | 0.02                    | 0.02                    | 0.02                    |
| Attention bias             | false                   | false                   | false                   |
| Attention dropout          | 0.0                     | 0.0                     | 0.0                     |
| Input BOS token ID         | 151643                  | 151643                  | 151643                  |
| Output EOS token ID        | 151645                  | 151645                  | 151645                  |
| Torch dtype                | bfloat16                | bfloat16                | bfloat16                |
| Use cache                  | true                    | true                    | true                    |
| Transformers version       | 4.51.0                  | 4.51.0                  | 4.51.0                  |

## Appendix F: DRL Training Summary

| Hyperparameter        | CSPI-LLM /<br>SAC  | SAC                | TD3                | DDPG               | TQC                |
|-----------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| Learning rate         | $3 \times 10^{-4}$ | $3 \times 10^{-4}$ | $3 \times 10^{-4}$ | $3 \times 10^{-4}$ | $3 \times 10^{-4}$ |
| Discount gamma        | 0.995              | 0.995              | 0.99               | 0.99               | 0.99               |
| Replay buffer         | 300,000            | 300,000            | 300,000            | 300,000            | 300,000            |
| Batch size            | 512                | 512                | 256                | 256                | 256                |
| Learning starts       | 50,000             | 50,000             | 50,000             | 50,000             | 50,000             |
| Target smoothing tau  | 0.005              | 0.005              | 0.005              | 0.005              | 0.005              |
| Entropy coefficient   | auto_0.05          | auto_0.05          | –                  | –                  | auto               |
| Target entropy        | –5.0               | –5.0               | –                  | –                  | auto               |
| Policy delay          | –                  | –                  | 2                  | –                  | –                  |
| Target policy noise   | –                  | –                  | 0.2                | –                  | –                  |
| Target noise clip     | –                  | –                  | 0.5                | –                  | –                  |
| Action noise std.     | –                  | –                  | 0.15               | 0.20               | 0.15               |
| Dropped quantiles     | –                  | –                  | –                  | –                  | 2                  |
| Batch norm            | –                  | –                  | –                  | –                  | –                  |
| Parallel environments | 32                 | –                  | –                  | –                  | –                  |
| Semantic interface    | enabled            | disabled           | disabled           | disabled           | disabled           |

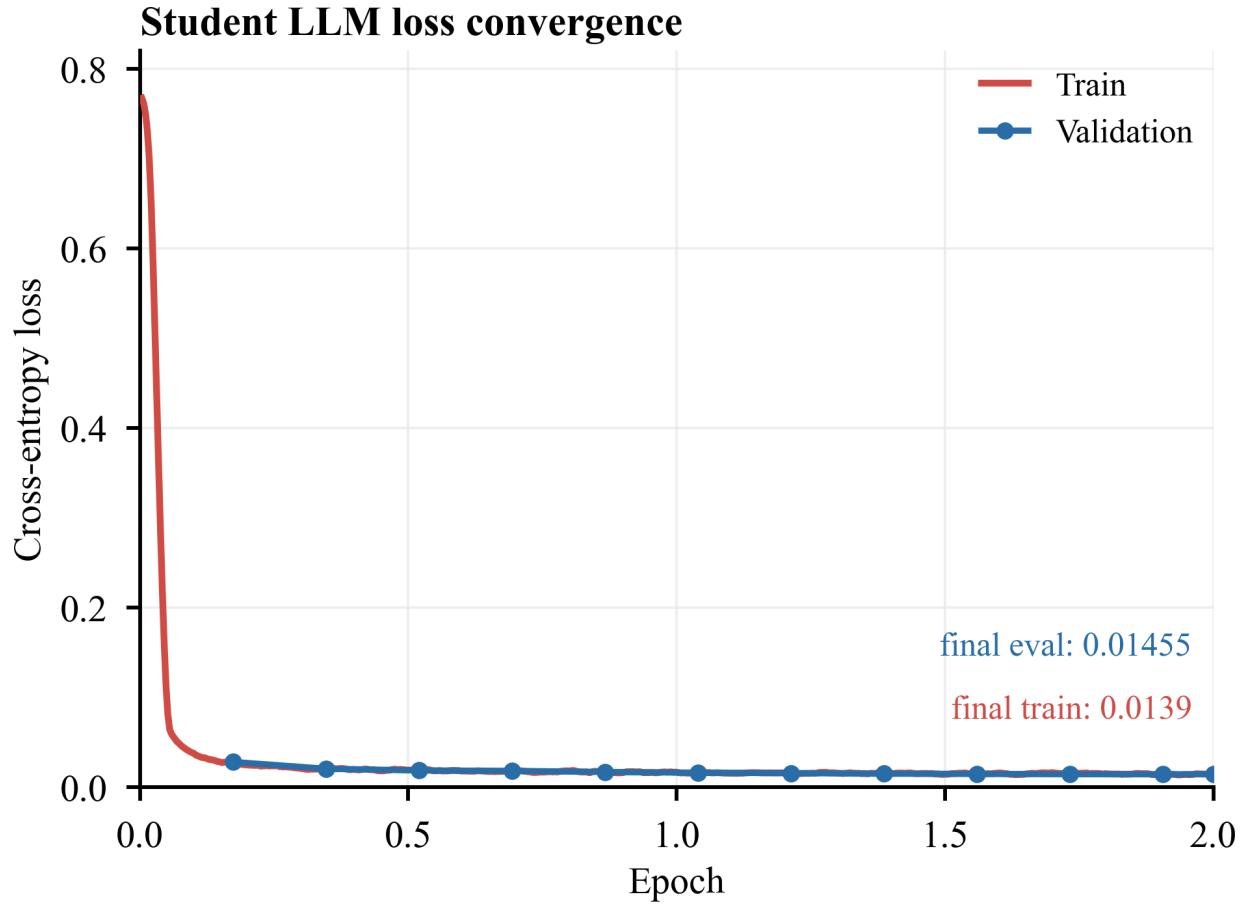
| Hyperparameter         | PPO                | TRPO               | Recurrent PPO      |
|------------------------|--------------------|--------------------|--------------------|
| Learning rate          | $3 \times 10^{-4}$ | $3 \times 10^{-4}$ | $3 \times 10^{-4}$ |
| Discount gamma         | 0.99               | 0.99               | 0.99               |
| Rollout steps          | 512                | 512                | 256                |
| Batch size             | 2,048              | 2,048              | 1,024              |
| Epochs                 | 5                  | –                  | 5                  |
| GAE lambda             | 0.95               | 0.95               | 0.95               |
| Clip range             | 0.2                | –                  | 0.2                |
| Entropy coefficient    | 0.01               | –                  | 0.01               |
| Value loss coefficient | 0.5                | –                  | 0.5                |
| Max grad norm          | 0.5                | –                  | 0.5                |
| TRPO target KL         | –                  | 0.01               | –                  |
| TRPO CG steps          | –                  | 15                 | –                  |
| LSTM hidden            | –                  | –                  | 256                |
| LSTM layers            | –                  | –                  | 1                  |

## Appendix G: Fig. 1



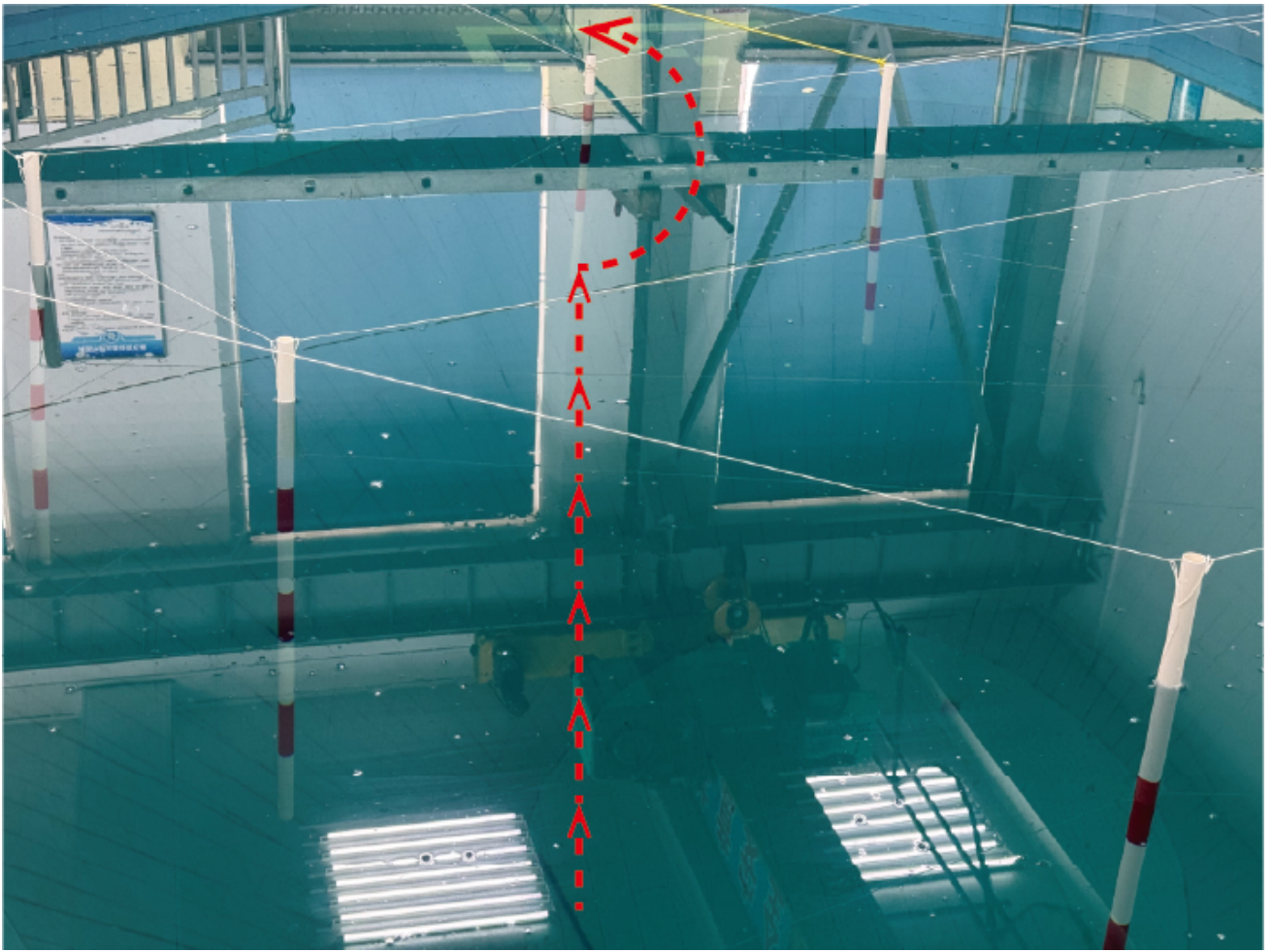
**Fig. 1.** Details of internal network components in CSPI-LLM. (a) Vision Transformer and multi-head attention blocks encode depth-derived visual observations. (b) The student LLM semantic vector and observation feature are fused through GateNet to form the final semantic-control representation. (c) The fused representation is used by the SAC actor-critic controller to generate the safety-fused action.

## Appendix H: Fig. 2



**Fig. 2.** Student LLM training-loss convergence. The curve reports the supervised fine-tuning loss recorded during LoRA adaptation on the filtered semantic dataset; the vertical axis gives the absolute loss scale used to assess convergence stability.

## Appendix I: Fig. 3



**Fig. 3.** Physical BSA-D2 water-tank experiment under the fixed thruster-generated-current condition. The dashed red curve indicates the ideal path, and the red-and-white frame marks the inspection structure used in the physical validation.



## Appendix J: Complete Request-Level Semantic Handoff Trace

This appendix reports the complete request-level semantic handoff trace used for the revised Table IV.

| $s_j$ (s) | $\ell_j$ (s) | $\kappa_j$ (s) | State       | $\Delta t_j^-$ (s) | Semantic vector     |
|-----------|--------------|----------------|-------------|--------------------|---------------------|
| 0.0       | –            | 0.0            | Commit      | 0.0                | [1 0 0 0 0 0 0 0 0] |
| 0.5       | 0.397        | 0.9            | Commit      | 0.9                | [1 1 0 0 0 0 0 0 0] |
| 1.0       | 0.537        | 1.6            | Commit      | 0.7                | [1 1 0 0 0 0 0 0 0] |
| 1.5       | 0.738        | 2.3            | Commit      | 0.7                | [1 1 0 0 0 0 0 0 0] |
| 2.0       | 0.516        | 2.6            | Commit      | 0.3                | [1 1 0 0 0 0 0 0 0] |
| 2.5       | 0.468        | 3.0            | Commit      | 0.4                | [0 1 0 0 0 0 0 0 0] |
| 3.0       | 0.728        | 3.8            | Commit      | 0.8                | [0 1 0 0 0 0 0 0 0] |
| 3.5       | 0.679        | 4.2            | Commit      | 0.4                | [0 1 0 0 0 0 0 0 0] |
| 4.0       | 0.576        | 4.6            | Commit      | 0.4                | [0 1 0 0 0 0 0 0 0] |
| 4.5       | 0.514        | 5.1            | Commit      | 0.5                | [0 1 0 0 0 0 0 0 0] |
| 5.0       | 0.682        | 5.7            | Commit      | 0.6                | [0 1 0 0 0 0 0 0 0] |
| 5.5       | 0.652        | 6.2            | Commit      | 0.5                | [0 1 0 0 0 0 0 0 0] |
| 6.0       | 0.595        | 6.6            | Commit      | 0.4                | [0 1 0 0 0 0 0 0 0] |
| 6.5       | 0.546        | 7.1            | Commit      | 0.5                | [0 1 0 1 0 1 0 0 0] |
| 7.0       | 0.596        | 7.6            | Commit      | 0.5                | [0 1 0 0 0 0 0 0 0] |
| 7.5       | 0.433        | 8.0            | Commit      | 0.4                | [0 1 0 1 0 1 0 0 0] |
| 8.0       | 0.471        | 8.5            | Commit      | 0.5                | [0 1 0 0 0 0 0 0 0] |
| 8.5       | 0.480        | 9.0            | Commit      | 0.5                | [0 1 0 0 0 0 0 0 0] |
| 9.0       | 0.503        | 9.6            | Commit      | 0.6                | [0 1 0 0 0 0 0 0 0] |
| 9.5       | 0.377        | 9.9            | Commit      | 0.3                | [0 1 0 0 0 0 0 0 0] |
| 10.0      | 0.831        | 10.9           | Hold→Commit | 1.0                | [1 1 0 0 0 0 0 0 0] |
| 10.5      | 0.579        | 11.1           | Commit      | 0.2                | [1 1 0 0 0 0 0 0 0] |
| 11.0      | 0.540        | 11.6           | Commit      | 0.5                | [0 1 0 0 0 0 0 0 0] |
| 11.5      | 0.389        | 11.9           | Commit      | 0.3                | [0 1 0 0 0 0 0 0 0] |
| 12.0      | 0.640        | 12.7           | Commit      | 0.8                | [0 1 0 0 0 0 0 0 0] |
| 12.5      | 0.536        | 13.1           | Commit      | 0.4                | [0 1 0 1 0 1 0 0 0] |
| 13.0      | 0.345        | 13.4           | Commit      | 0.3                | [1 1 0 0 0 0 0 0 0] |
| 13.5      | 0.584        | 14.1           | Commit      | 0.7                | [0 1 0 0 0 0 0 0 0] |
| 14.0      | 0.393        | 14.4           | Commit      | 0.3                | [0 1 0 0 0 0 0 0 0] |
| 14.5      | 0.362        | 14.9           | Commit      | 0.5                | [1 1 0 0 0 0 0 0 0] |
| 15.0      | 0.363        | 15.4           | Commit      | 0.5                | [0 1 0 0 0 0 0 0 0] |
| 15.5      | 0.828        | 16.4           | Hold→Commit | 1.0                | [0 1 0 1 0 1 0 0 0] |
| 16.0      | 0.680        | 16.7           | Commit      | 0.3                | [0 1 0 0 0 0 0 0 0] |
| 16.5      | 0.614        | 17.2           | Commit      | 0.5                | [0 1 0 0 0 0 0 0 0] |
| 17.0      | 0.694        | 17.7           | Commit      | 0.5                | [0 1 0 1 0 1 0 0 0] |
| 17.5      | 0.777        | 18.3           | Commit      | 0.6                | [0 1 0 0 0 0 0 0 0] |
| 18.0      | 0.541        | 18.6           | Commit      | 0.3                | [0 1 0 0 0 0 0 0 0] |
| 18.5      | 0.337        | 18.9           | Commit      | 0.3                | [0 1 0 0 0 0 0 0 0] |
| 19.0      | 0.849        | 19.9           | Hold→Commit | 1.0                | [0 1 0 0 0 0 0 0 0] |
| 19.5      | 0.769        | 20.3           | Commit      | 0.4                | [0 1 0 0 0 0 0 0 0] |
| 20.0      | 0.749        | 20.8           | Commit      | 0.5                | [0 1 0 0 0 0 0 0 0] |
| 20.5      | 0.743        | 21.3           | Commit      | 0.5                | [0 1 0 0 0 0 0 0 0] |
| 21.0      | 0.617        | 21.7           | Commit      | 0.4                | [0 1 0 0 0 0 0 0 0] |
| 21.5      | 0.645        | 22.2           | Commit      | 0.5                | [1 1 0 0 0 0 0 0 0] |
| 22.0      | 0.386        | 22.4           | Commit      | 0.2                | [0 1 0 0 0 0 0 0 0] |
| 22.5      | 0.447        | 23.0           | Commit      | 0.6                | [0 1 0 0 0 0 0 0 0] |
| 23.0      | 0.658        | 23.7           | Commit      | 0.7                | [0 1 0 0 0 0 0 0 0] |
| 23.5      | 0.332        | 23.9           | Commit      | 0.2                | [1 1 0 0 0 0 0 0 0] |
| 24.0      | 0.872        | 24.9           | Hold→Commit | 1.0                | [1 1 0 0 0 0 0 0 0] |
| 24.5      | 0.432        | 25.0           | Commit      | 0.1                | [1 1 0 0 0 0 0 0 0] |

## Appendix K: Evaluation Metrics

All six metrics are computed from the logged state, action, and clearance histories over each evaluation episode.

1. **Path efficiency (%)**:  $PE = \frac{L_{\text{route}}}{L_{\text{traveled}}} \times 100$ , where  $L_{\text{route}}$  is the length of the nominal route and  $L_{\text{traveled}} = \sum_t \|\mathbf{p}_{t+1} - \mathbf{p}_t\|_2$  is the traveled distance.
2. **Minimum clearance (m)**:  $MC = \min_t \min_i d_{\text{obs},i}(\mathbf{p}_t)$ , the minimum distance to the nearest obstacle over all control instants of the episode.
3. **Risk-time ratio (%)**:  $RTR = \frac{1}{T} \sum_t \mathbf{1}\{MC_t < d_{\text{risk}}\} \times 100$ , the fraction of control instants at which the minimum clearance falls below the risk distance  $d_{\text{risk}} = 2.25$  m (the conservative stopping-distance estimate of the platform at the evaluated operating speed).
4. **Action energy**:  $AE = \frac{1}{T} \sum_t \|\mathbf{a}_t\|_2^2$ , the mean squared action magnitude over the episode.
5. **Action smoothness**:  $AS = \frac{1}{T} \sum_t \|\mathbf{a}_t - \mathbf{a}_{t-1}\|_2$ , the mean norm of consecutive action differences.
6. **Path length (m)**:  $L_{\text{traveled}} = \sum_t \|\mathbf{p}_{t+1} - \mathbf{p}_t\|_2$ , the total traveled distance.